

Summary

House Price Prediction

This project aimed to predict house prices using machine learning techniques and property-related features from the Housing dataset. After cleaning the data, handling categorical variables, and removing duplicates, two regression models—Linear Regression and Random Forest—were trained and evaluated. The dataset was explored, cleaned, and preprocessed by handling missing values, removing duplicate records, and converting categorical variables into numerical format using one-hot encoding.

Two regression models, Linear Regression and Random Forest Regressor, were trained and evaluated using an 80:20 train-test split. Model performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R^2 Score.

In accuracy among the models tested, the Linear Regression provided better performance, achieving a higher R^2 Score of 0.653 which means it was able to explain about 65% of the variation in house prices. It also produced lower prediction errors (MAE $\approx$ 970,000 and RMSE $\approx$ 1,324,000) compared to the Random Forest model. Key factors influencing house prices included area, number of bathrooms, furnishing status, air conditioning, and preferred location. It was also observed that houses with premium amenities tended to have higher prices, even when their sizes were similar.

Visualizations such as a price distribution histogram, correlation heatmap, and actual vs. predicted scatter plot provided useful insights into the dataset.

Based on these findings, real estate businesses can benefit from using data-driven pricing models to estimate property values more accurately and support informed decision-making. Such approaches can help buyers, sellers, and agents make informed decisions, improve pricing strategies, and enhance overall market efficiency.

By :

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